



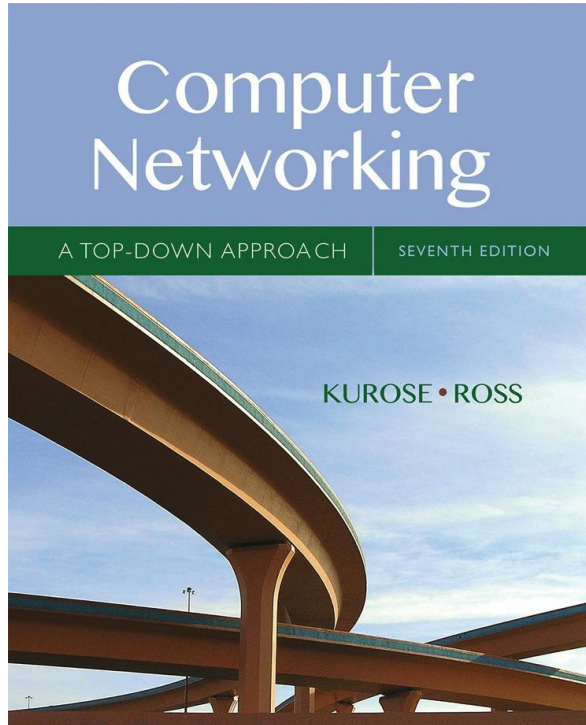
COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

Text Book



Computer Networking: A Top-Down Approach

Jim Kurose, Keith Ross
Pearson, 2017
7th Edition

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Unit – 1 Computer Networks and the Internet

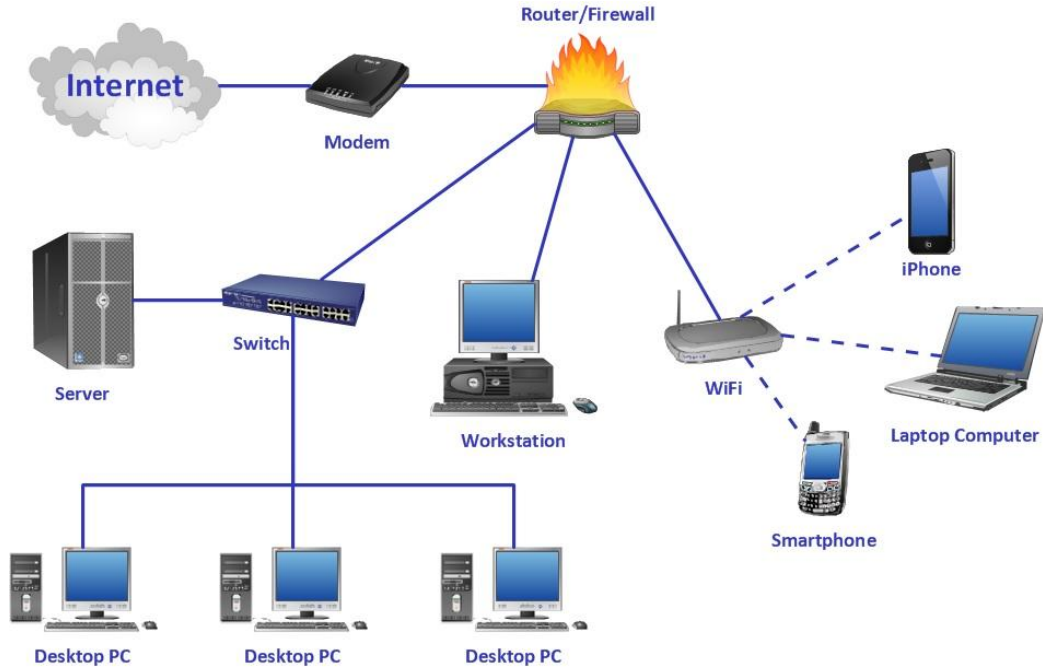
1.1 Introduction to Computer Networks

1.2 What is the Internet?

- A nuts-and-bolts and Services description, Protocol

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Introduction to Computer Networks



- Two or more devices connected together.
- Communicate with each other, share data or resources

- A massive network of networks.
- A computer network that interconnects billions of computing devices throughout the world.
- Traditional devices – PCs, Workstations, Servers – web pages, emails, etc.
- Internet “things” – laptops, PDAs, TVs, gaming consoles, home security systems, home appliances, watches, cars, traffic control systems, etc.,

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The Internet: A “Nuts and Bolts” View



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

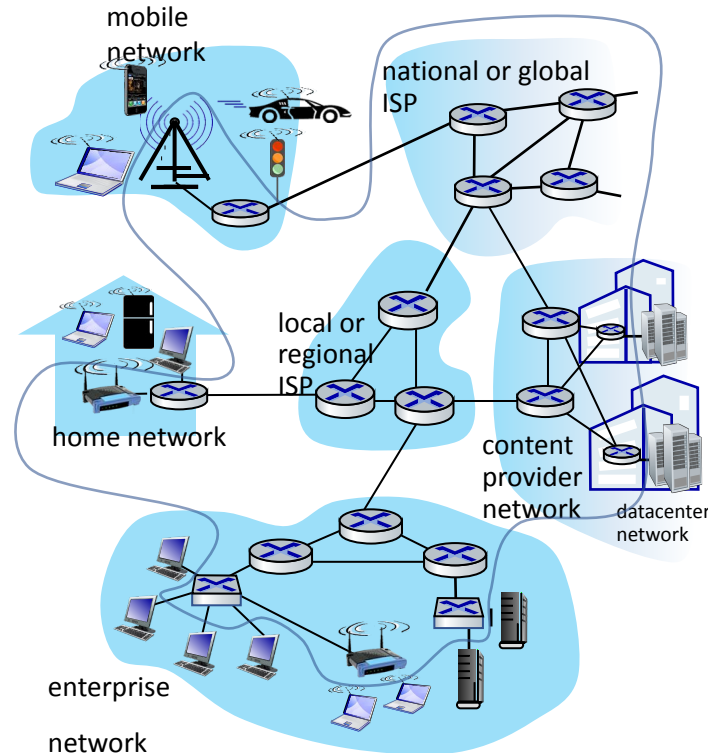


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



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"Fun" Inter-connected Devices



Amazon Echo



Internet refrigerator



IP picture frame



Tweet-a-watt:
monitor energy
use



Web-enabled toaster +
weather forecaster



Security Camera



Slingbox: remote
control cable TV



AR devices

Internet phones



sensorized,
bed
mattress



Fitbit

Others?

**There will be 41 Billion
IoT devices by 2027***

* <https://www.businessinsider.com/internet-of-things-report?IR=T>

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The Internet: A “Nuts and Bolts” View

- *Internet: “network of networks”*

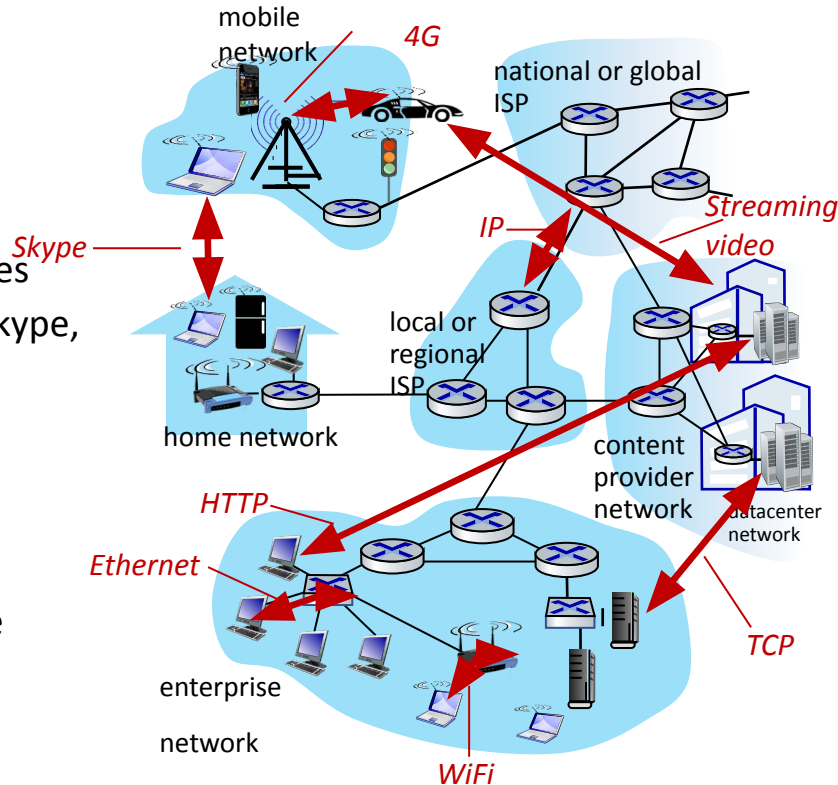
- Interconnected ISPs

- *Protocols are everywhere*

- control sending, receiving of messages
- e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet

- *Internet standards*

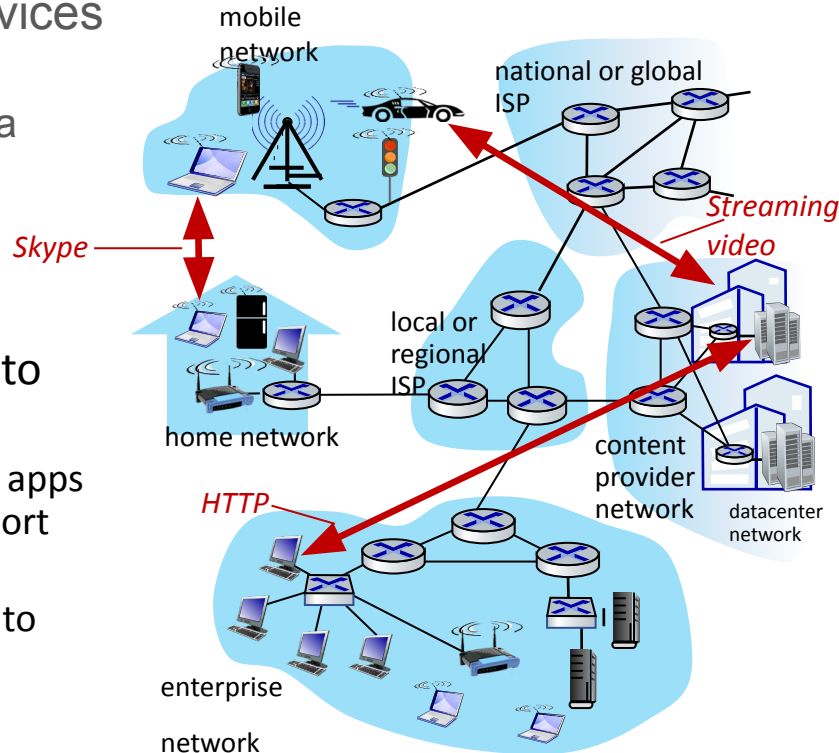
- RFC: Request for Comments
- IETF: Internet Engineering Task Force



- **Infrastructure** that provides services to applications:

- Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...

- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific messages sent

... specific actions taken when
message received, or other
events

Network protocols:

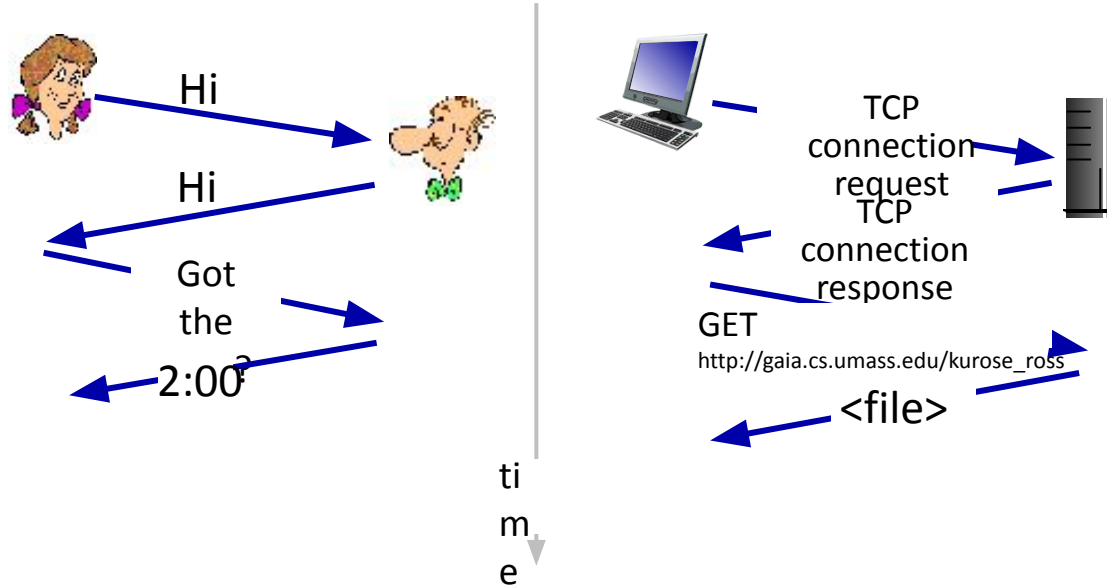
- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on msg transmission, receipt.*

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What's a Protocol?

A human protocol and a computer network protocol:



Q: other human protocols?



THANK YOU

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